Concept of LOSS Function

$$Let\ \underline{\mathbf{Y}} = \begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{pmatrix}_{N \times 1}$$

Original Labels

from Training Samples

$$Let\ \underline{\hat{\mathbf{Y}}} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \dots \\ \hat{y}_n \end{pmatrix}_{N \times 1}$$

Predicted Labels

from Model $\hat{Y}=X\underline{W}$

Remember

$$\|a\|_2 = a^T a$$

Objective is $\hat{Y}=Y$

If $\hat{Y} \neq Y \Rightarrow$ There is an **Error (Loss)**

$\underline{\hat{\mathbf{Y}}} = \boldsymbol{X}\ \underline{\boldsymbol{W}}$ ***Predicted***

Where : $\boldsymbol{X}_{N \times d+1}$ is a **matrix** $\Rightarrow$ **Given**

$\underline{\boldsymbol{Y}}_{N \times 1}$ is labels **vector** $\Rightarrow$ **Given**

$\underline{\boldsymbol{W}}$ is Model Parameter $\Rightarrow$ **Required**

Define **Loss** function as **squared distance** between $\hat{Y}$ and Y

$$Loss = \|Y - \hat{\mathbf{Y}}\|_2 = (Y - \hat{\mathbf{Y}})^T (Y - \hat{\mathbf{Y}})$$